Chapter 2 · Story

A brighter poster. More visitors. A cause?

An original story at the Common Ground repair café.



The poster that “worked” · c02s01

A fictional story. A real question about people.

What would you do?

Which statement fits Leo’s evidence?

A The poster caused the increase.

B The poster had no effect.

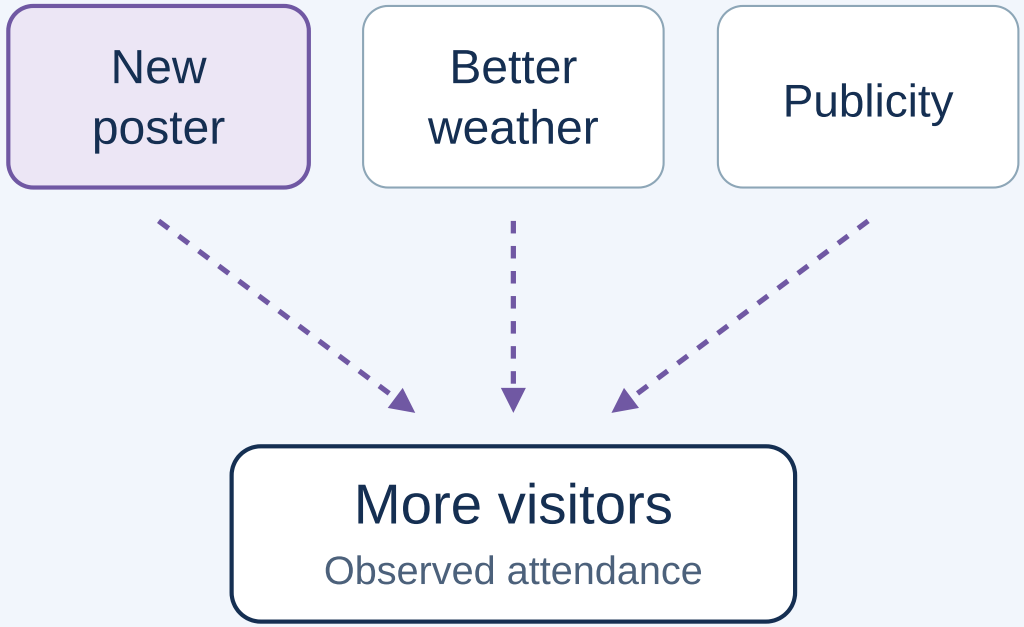
C Attendance rose; several causes remain possible.

Suggested answer: C. Saying that attendance rose preserves the observation. Leaving the cause open reflects the limits of the comparison.

Comparison makes a claim testable.

- **Correlation**: things vary together.
- **Confound**: another factor could explain the pattern.
- **Random assignment**: allocate conditions by chance.
- **Random sampling**: select people from a population by chance.

Several changes, one outcome



Dashed paths: possible explanations

A fair comparison separates explanations.

Several changes can fit the same before-and-after pattern.

Ask how much—and how uncertain.

- **Size**: 12% versus 15% is a difference of three percentage points.
- **Uncertainty**: another sample may give a different estimate.
- **Statistical significance**: judged against a model, often assuming no group difference.
- Significance does not establish practical importance; nonsignificance does not prove zero effect.

Invented attendance example

Invitation A



12 of 100 households attend

Invitation B



15 of 100 households attend

15% − 12% = 3 percentage points

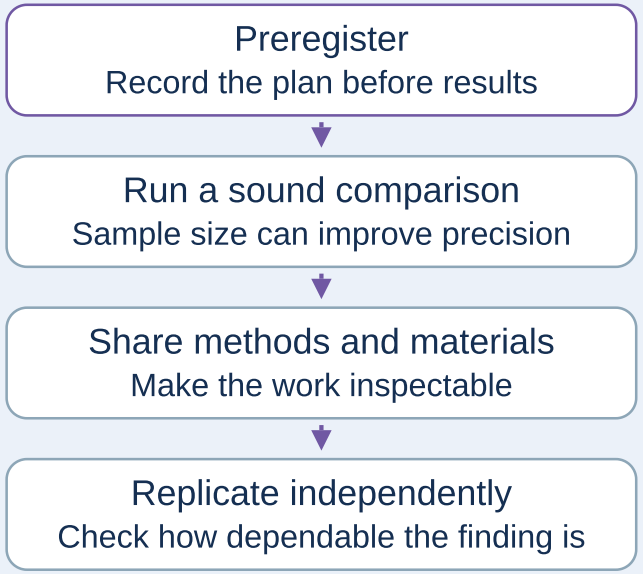
A different sample could give a different estimate.

Invented counts illustrate the size of a difference.

What can we really conclude?

Replication repeats a test to examine how dependable a finding is. Independent tests help us learn whether a result holds up elsewhere. Preregistration records the planned methods and analysis before results are known. That advance record reduces some opportunities for selective reporting. Preregistration cannot rescue a poor design. With a sound design, a larger sample can narrow the uncertainty around an estimate. More observations cannot fix a systematically unfair comparison. Sharing the study's methods and materials lets others inspect how it was done. A memorable study is a starting point for evaluation, rather than a final verdict.

Build confidence through checks



A larger sample cannot fix a biased comparison.

Plan, inspect, and independently repeat a sound comparison.

Try the idea somewhere new.

Which invitation comparison is strongest?

A Design A in winter; design B in summer.

B Randomize designs for one event; define attendance in advance.

C Design A for existing fans; design B for newcomers.

Suggested answer: B. Randomizing invitations for one event improves the comparison. Also check whether households share invitations.

Close your notes. Explain it.

> How does random assignment differ from random sampling?

> Why is before-and-after evidence insufficient?

1. Random assignment allocates people to conditions. Random sampling selects people from a population.

2. Other conditions can change over time and produce the same before-and-after pattern.



Maya, Sam and Leo at the repair café.

Carry one idea into your week.

Find an everyday causal claim. Name its outcome, one competing explanation, and a feasible comparison. State whose behavior the result would represent.

Revisit tomorrow and in a week.
Explain it without notes, then use a different example.



Take the question into a different setting.